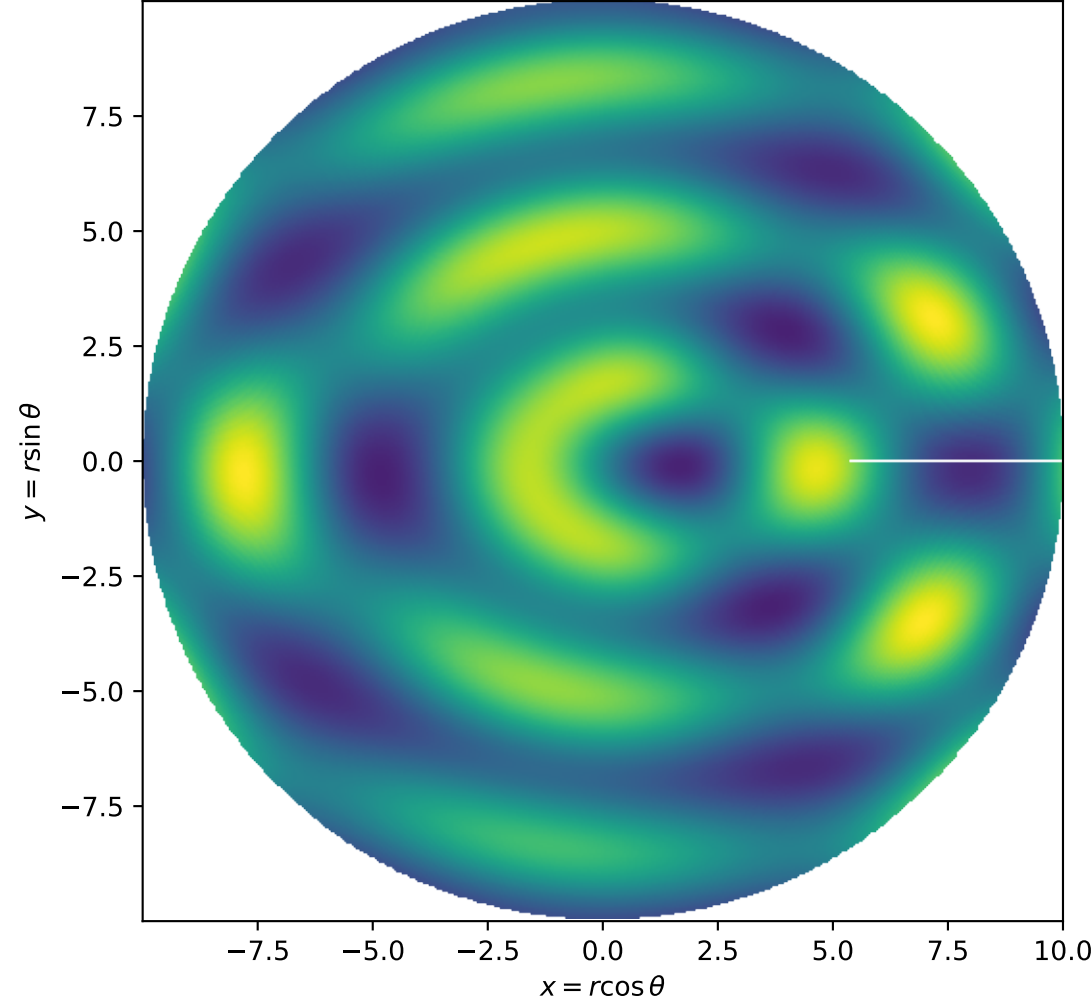


AI-Assisted Ansatz Discovery for Nonlinear Pattern Formation

5 × 5 validation MSE over (μ, ν)

average MSE ≈ 45.00

Discovered ansatz pattern at μ = ν = 5
SH residual MSE ≈ 55.50



Steady Swift–Hohenberg equation

$$-(1 + \Delta)^2 f + \mu f + \nu f^2 - f^3 = 0$$

$$\Delta f = f_{rr} + \frac{1}{r} f_r + \frac{1}{r^2} f_{\theta\theta}$$

Domain and constraints

$$r \in [0.01, 10], \quad \theta \in [0, 2\pi), \quad \mu, \nu \in [0.01, 10]$$

periodic in θ ; nontriviality enforced by spatial-derivative gates

Symbolic regression ansatz

$$f(r, \theta; \mu, \nu) \approx B(r, \theta, \mu, \nu)$$

$$- A(r, \mu, \nu) \sin(r) \sin(\Phi(\mu, \nu) - \alpha r \sin \theta - \theta + C)$$

Current expression: 75 DAG nodes; SH residual DAG: 493 nodes.